Jiayi Wang (James)

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EDUCATION

Georgia Institute of Technology, Atlanta, GA *August 2025 - Present*
M.S. in Music Technology; GPA 4/4; Music Informatics Lab; Robotic Musicianship Lab

Occidental College, Los Angeles, CA *August 2020 - May 2025*
B.A. in Music, Emphasis in Music Production; B.A. in Computer Science; Minor in Mathematics
GPA 3.88/4; **Magna Cum Laude**; Dean's List: all semesters
Kurtich Music Grant: four semesters of scholarship from 2023 to 2025

PUBLICATIONS & ACADEMIC SERVICE

- A. T. Ma, A. Cammarota, **J. Wang**, A. Smith, C. Z. A. Huang, J. Albert, and A. Lerch, "H2H Music Improv: A Communication Model and Audio-Visual Dataset for Music Improvisation," Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), 2026.
- S. Hudson and **J. Wang**, "Chord Tone or Harmonic-Bass Divorce? An Enactive Approach to Hearing Pedals in Popular Music," Music Theory Online, vol. 31, no. 4, 2025.
- **J. Wang**, "Cinematic Pedal in Rock and Pop Music," presented at the Occidental College Summer Research Conference, Los Angeles, CA, Aug. 2023.
- **J. Wang**, "Re-evaluating Use Cases in VIVO," presented at the Andrew W. Mellon Summer Faculty Institute Conference, Los Angeles, CA, Aug. 2022.
- Reviewer, International Society for Music Information Retrieval Conference (ISMIR), 2026.

INDUSTRY EXPERIENCE

Audio Research Intern, Tencent Music Entertainment (TME), Guangzhou, China *June - August 2026*
- Developed a MIDI-to-guitar fingering **LLM** model; fine-tuned **T5** and **Qwen** on 140k Guitar Pro files; achieved **84.47% TAB accuracy** and **99.96% pitch consistency** on a 14,368-track test set containing 15.1 million notes.
- Engineered an end-to-end guitar accompaniment TAB generation pipeline that transforms structured musical data into performance-ready MusicXML scores; generated **528 arrangements for 176 songs across three difficulty levels**, with **148 passing tests** and zero export failures.
- Built and deployed a password-protected **internal full-stack platform** for score review and version management, supporting drag-and-drop file/folder uploads, deletion, and distribution of more than **1,400 generated score files**.

Data Science Intern, Leizhi Information Technology, Shanghai, China *July – August 2024*
- Participated in **data analysis of Geographic High-Risk Merchants** for a government agency in Pudong, Shanghai; developed **GIS-based geographic heat maps** to support government regulatory oversight.
- Participated in the development of a **real-time complaint and reporting monitoring system** for a government agency, providing decision support for government management.

Audio Interface Designer, Huaxin Musical Instrument, Los Angeles, CA *January - May 2024*
- Designed **a soundboard-style audio interface** in Blender for live streaming and recording with electronic piano and drum kits; specified three preamp channels, XLR and line inputs, VST-compatible audio drivers, multiple sample rates and bit depths, USB-C connectivity, and electrical parameters optimized for performance and affordability.

RESEARCH EXPERIENCE

ICASSP 2027 Submission, Optical Music Recognition for Jazz Lead Sheet: GitHub *August 2025 - Present*

- Proposed an end-to-end **system for full-page handwritten jazz lead sheet recognition**, leveraging a Sheet Music Transformer to jointly transcribe melody and chord symbols from images.
- Designed a **9-stage curriculum learning strategy** to progressively scale from single-staff inputs to full-page transcription using ~1.8k samples.
- Developed a detect-and-concatenate baseline using YOLO-based staff detection followed by staff-level recognition and reconstruction, and introduced chord-focused evaluation metrics.
- Achieved **substantial improvements** on a 32-page handwritten test set, reducing chord symbol error rate from 47.8% to 30.7% ($p < 0.001$), significantly outperforming the baseline.

Independent Research, Real-Time Human-Robot Improvisation System: GitHub *January 2026 - Present*

- Developed a **real-time human–robot improvisation system** based on the Anticipatory Music Transformer (AMT), extending it from offline MIDI continuation to interactive co-performance with a marimba-playing robot.
- Fine-tuned AMT on the PiJAMa jazz improvisation MIDI dataset to better align generated output with jazz improvisational style and human musical interaction.
- Designed a real-time pipeline where human MIDI input is streamed via OSC to a cloud GPU for inference and Viterbi decoding, then transmitted back for robotic actuation.

Audio Content Analysis Project, Salient Detection for F0 Estimation; GitHub, PDF *September - December 2025*

- Built a **deep learning system** for fundamental frequency estimation, predicting melody from audio signals.
- Designed a graphical audio representation (**HCQT**) and trained a **convolutional neural network** for F0 prediction.
- Conducted controlled comparisons between rule-based methods, CNNs, and transfer learning from ResNet18; the CNN achieved **81.6% overall accuracy** and demonstrated the importance of domain-specific inductive bias.

Audio Content Analysis Class Project, Audio Patterns for Prediction; GitHub, PDF *September - December 2025*

- Implemented an **audio-to-audio generation system** that predicts melody continuations from raw audio with a data processing pipeline, including tempo normalization, spectrogram slicing, feature extraction, and clustering.
- Implemented and compared multiple temporal models (**HMM variants, including a self-designed variable-order HMM**) to analyze trade-offs between short- and long-range contextual conditioning.
- Evaluated generated outputs using standardized MIREX metrics (Cardinality and Pitch Scores), achieving performance **comparable** to symbolic sequence baselines.

Software Development, Gesture Ensemble Music System (GEMS): GitHub, PDF *February 2024 - Present*

- Designed and implemented **GEMS**, a gesture-controlled human–machine improvisation system enabling a solo performer to interact with a responsive virtual ensemble in **real time** using Max/MSP.
- Built **genre-adaptive generative engines** (chords, bass, drums) driven by symbolic chord progressions, providing harmonically grounded accompaniment for improvisation practice.
- Implemented real-time **gestural control** of music generation using MediaPipe-based pose tracking, streaming performer movement data via **UDP** to influence generative behavior and ensemble interaction.

Undergraduate Thesis, Music Dept. Senior Comprehensive Project: GitHub, PDF *June - December 2024*

Tracing the Evolution of Rock and Popular Music Harmony: A Computational Study Using Shannon Entropy

- Designed a computational musicology study of 890 rock songs (1950s–2020s) from the McGill Billboard dataset.
- **Implemented Shannon entropy** as a harmonic complexity metric, demonstrating the viability of information theory as a quantitative framework for genre evolution research.
- **Discovered a measurable rise and fall in harmonic complexity**, peaking in late-1970s progressive rock before declining in the 1980s and diversifying through the 1990s–2000s.

MUSIC STUDIO INDUSTRY EXPERIENCE

Production Manager Assistant, Occidental College Music Department, Los Angeles, CA *January 2022 – May 2025*

- Assisted with event promotion, logistics, and social media management for on-campus live musical performances (15-20 events per semester).
- Livestreamed and recorded events on YouTube; set up and dismantled livestream equipment and monitored the soundboard, OBS streaming system, and other audio equipment.
- Managed inventory (cameras, microphones, speakers, soundboards, cables), created a labeling system for equipment, and maintained a digitized archive of past performance materials.

Studio Assistant, Occidental College Choi Studio, Los Angeles, CA *August 2023 – May 2025*

- Managed weekly inventory (worth $60K+), checked equipment in and out, and monitored its condition after returns; maintained studio and storage spaces and installed software to ensure readiness for use.
- Supervised and provided technical support for students and enforced the studio's leasing policies (2-3 three-hour sessions per week).
- Designed and managed the Electronic Music Studio (EMS) at Occidental College, including installing audio equipment, managing software updates, and coordinating work orders across Music, IT, and Maintenance.

Recording Intern, Kingsize Soundlabs, Los Angeles, CA *February – June 2023*

- Shadowed professional engineers in recording sessions, including collaborations with Lyla Forde, Will Brahm, Walter Trout, Aidan Carroll, Julius Rodriguez, Drown, and Dudley.
- Assisted in studio operations, including assembling and disassembling recording equipment, daily studio cleanup and maintenance, managing inventories, maintaining the single VIP vocal studio, and coffee and lunch runs.

LEADERSHIP & SERVICE

- **Vice President**, Georgia Tech Audio Engineering Society (AES) Club — Organize mix-feedback sessions, train students to use Georgia Tech's recording studios and audio equipment, and coordinate guest lectures with industry professionals and researchers.
- **Treasurer**, Occidental Chinese Students & Scholars Association — Managed an annual budget of **$5,000** and organized Lunar New Year celebrations, biweekly movie nights, concerts, bonfires, and new-student social events.
- **Student Representative**, Suno AI at Occidental College

SKILLS & INTERESTS

- Programming: Java, Python, C/C++, HTML/CSS, Max/MSP, SQL, Blender, LaTeX, Mathematica
- Music production: DAWs (Logic Pro X, Ableton Live, Pro Tools), iZotope RX, iZotope Ozone, Melodyne
- Instruments: Saxophone (10+ years), Guitar, Piano, Tuba, and Drums
- Languages: Mandarin Chinese (native) and English (fluent)
- Interests: esports (League of Legends North America top 20%), Oxy Intramural Basketball Team, Snowboarding (black diamond, freestyle park)